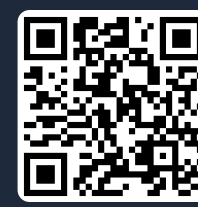




THEMIS: Unlocking Pretrained Knowledge with Foundation Model Embeddings for Anomaly Detection in Time Series

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1. Introduction & Key Contributions

The Challenge

Time series anomaly detection is critical but challenging due to data complexity, noise, and the high cost of manual tuning. Existing methods often lack the flexibility and generalizability needed for diverse, real-world scenarios.

Our Solution: THEMIS

We introduce **THEMIS**, a novel, **zero-shot** framework that leverages the rich, pretrained knowledge from Time Series Foundation Models (TSFMs) like Chronos to perform robust and interpretable anomaly detection.

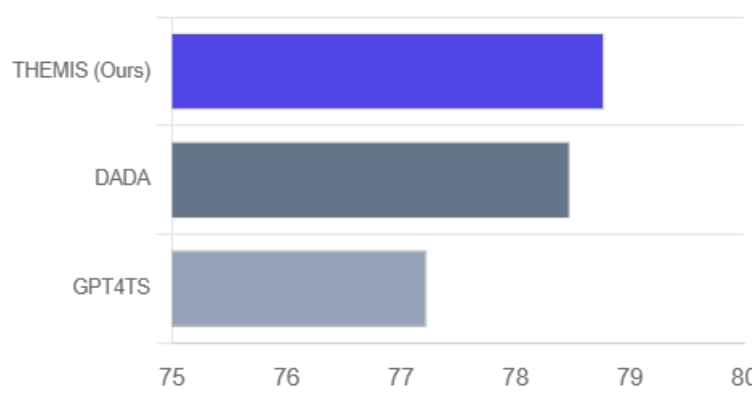
Key Contributions

- Novel Zero-Shot Framework:** We successfully decouple representation learning from anomaly scoring, using frozen foundation model embeddings for detection without any task-specific training.
- State-of-the-Art Performance:** THEMIS achieves a new SOTA F1-score of **78.78%** on the MSL benchmark, outperforming specialized models.
- Practical and Robust:** The framework is highly resilient to hyperparameter changes, making it a practical and reliable solution for real-world deployment where labeled data is scarce.

3. Results & Analysis

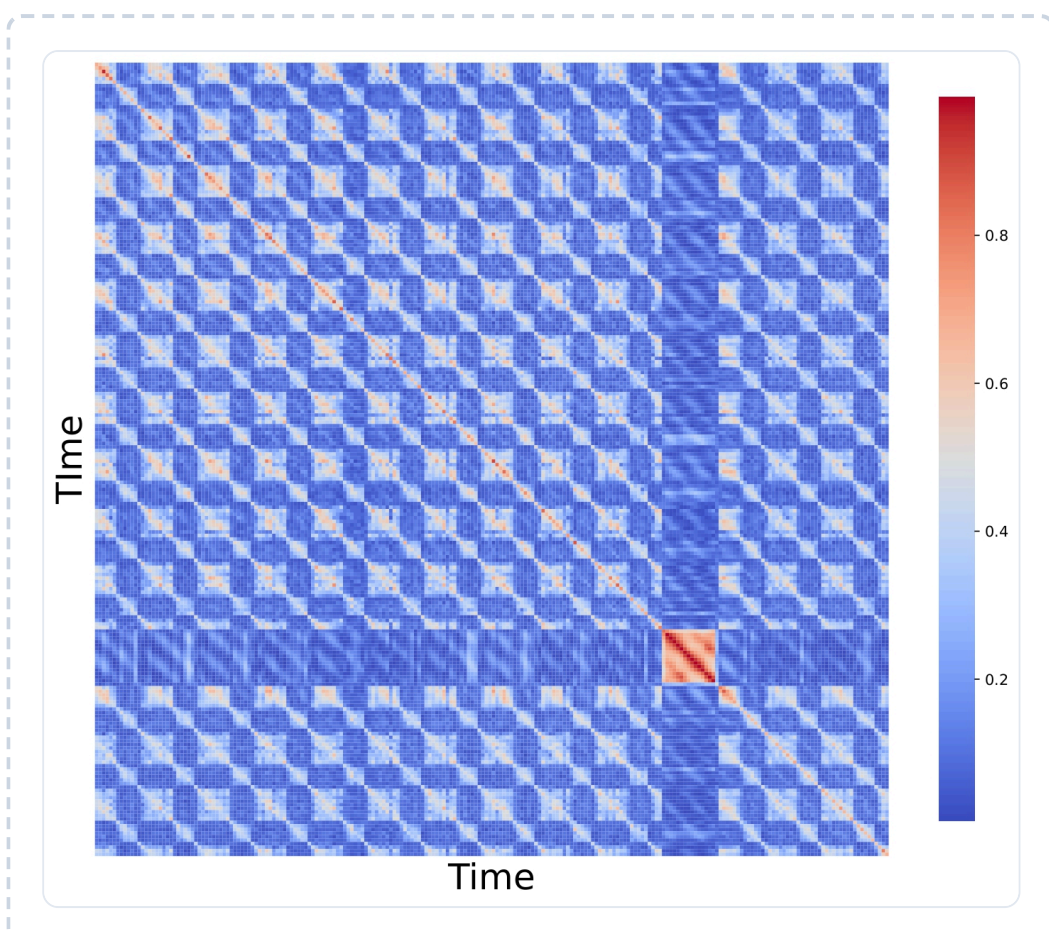
State-of-the-Art on MSL Benchmark

THEMIS outperforms leading models, demonstrating the power of leveraging pretrained knowledge.



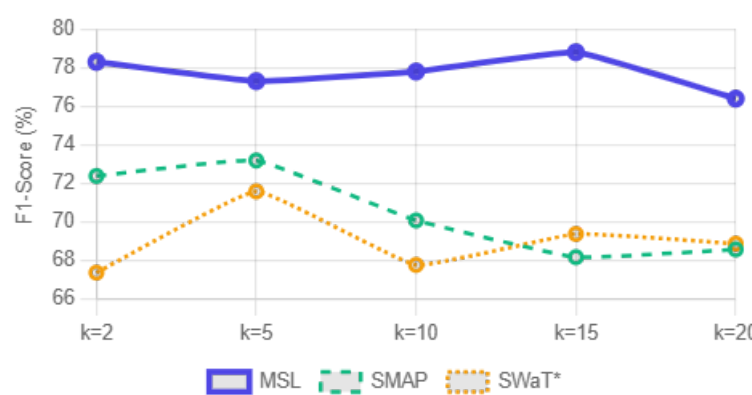
Visual Analysis: Anomaly Detection in Action

THEMIS accurately identifies anomalous intervals by assigning them high scores, aligning perfectly with ground-truth data from the SMAPE dataset.

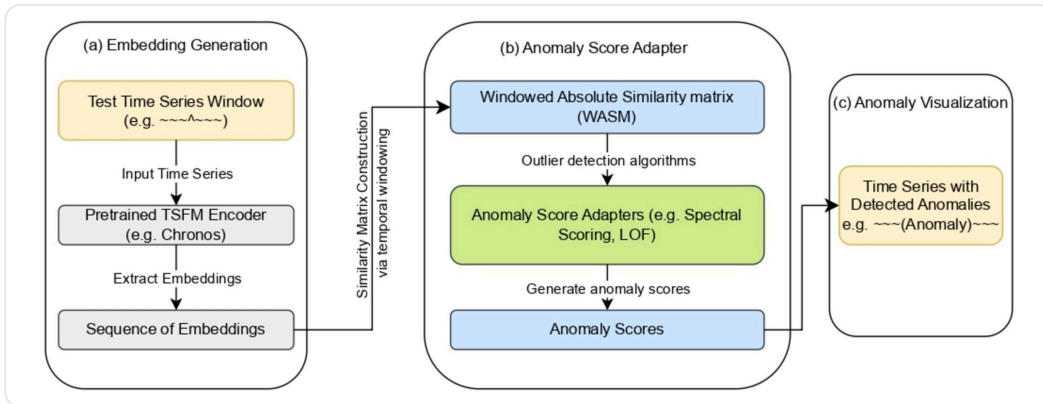


Hyperparameter Robustness

The framework's performance is remarkably stable across a wide range of hyperparameter choices, making it highly practical.



2. Methodology: How THEMIS Works



A Three-Step Process

1. Embedding Generation

A time series is segmented into sliding windows. The frozen **Chronos** encoder transforms each window into a sequence of rich, contextual embeddings.

2. Similarity Matrix Construction

We compute the **Windowed Absolute Similarity Matrix (WASM)** from these embeddings to capture the underlying structural patterns of the time series segment.

$$S[i,j] = | \langle z_i, z_j \rangle | / (\|z_i\|_2 \cdot \|z_j\|_2)$$

3. Anomaly Scoring

A score adapter is applied to the WASM to identify deviations. Our primary and best-performing method is **Spectral Residual Scoring**, which identifies points that do not align with the dominant structural patterns of the series. Firstly, we perform an eigendecomposition on the WASM:

$$S = Q \Lambda Q^T$$

Then, we use the top-k eigenvectors ($e_1, e_2, \dots, e_k$) to form a spectral embedding. The anomaly score s_t for a point t is its normalized distance from this dominant subspace:

$$\text{Anomaly Score } (s_t) = 1 - (\|e_t\|_2 / \max_i \|e_i\|_2)$$

4. Conclusion & Future Work

Conclusion

THEMIS is a powerful demonstration that general-purpose embeddings from TSFMs provide a robust foundation for anomaly detection. By combining these rich representations with global, structure-aware scoring methods, our zero-shot framework achieves state-of-the-art results with exceptional practicality.

Future Directions

- Adapt the framework for **multivariate** time series.
- Explore embeddings from other advanced **TSFMs**.
- Investigate a wider range of sophisticated **outlier detection algorithms**.

References

- Ansari, A. Z., et al. (2024). Chronos: Learning the language of time series. arXiv preprint arXiv:2403.07815.
- Ren, J., et al. (2019). Time-series anomaly detection service at microsoft. In Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining.
- Su, Y., et al. (2019). Robust anomaly detection for multivariate time series through stochastic recurrent neural network. In Proceedings of the 25th ACM SIGKDD.
- Li, Z., et al. (2021). DADA: Devirt and Denoise for Anomaly Detection in Time Series. arXiv preprint arXiv:2106.06941.
- Zhou, B., et al. (2023). GPT4TS: A new baseline for time series forecasting. arXiv preprint arXiv:2305.02529.